

MIN Function in Excel

1. What is MIN?

The MIN function returns the smallest number in a set of values. It solves the problem of finding the lowest value in a long list, such as the lowest sale, the lowest score, or the cheapest price. A Data Analyst uses it to find the weakest results, the best (lowest) costs, and the starting point of a range of values.

2. Syntax

`=MIN(number1, [number2], ...)`

Argument	Required?	What it accepts
number1	Required	A number, a cell reference, or a range (for example B2:B5)
number2, ...	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=MIN(B2:B5)`. You can also mix a range with a number: `=MIN(B2:B5, 20000)` returns whichever is smaller, the lowest value in the range or 20000.

3. Simple Example

Find the lowest sales figure:

	A	B
1	Employee	Sales
2	Rahim	25000
3	Karim	42000
4	Hasan	35000
5	Nabil	50000
6	Minimum	25000

Formula in B6: `=MIN(B2:B5)`

Excel compares the four values in B2 to B5 (25000, 42000, 35000, 50000) and returns the smallest one. **Result: 25000**

4. Another Example

Find the lowest price. The Monitor price is not decided yet:

	A	B
1	Product	Price
2	Mouse	25
3	Keyboard	45
4	Monitor	TBD
5	Headset	18
6	Webcam	60
7	Lowest	18

Formula in B7: `=MIN(B2:B6)`

Excel ignores the text "TBD" and compares only 25, 45, 18, and 60. **Result: 18**

5. When to Use It

- Find the lowest sales figure in a period
- Find the lowest exam score to see who needs support
- Find the lowest salary in a department
- Find the cheapest price or lowest cost, such as the best supplier quote
- Find the earliest date in a list, such as the first order date (dates are stored as numbers)

6. Important Things to Remember

- Text and empty cells inside a range are ignored. They are not counted as zero.
- A cell containing 0 is counted, so a 0 entered for missing data becomes the minimum. Among negative numbers, the smallest is the one furthest from zero, so -20 is smaller than -5.
- If a referenced cell contains an error such as #N/A, the whole MIN result shows that error.
- Numbers stored as text (often left-aligned) are ignored in a range. Convert them to real numbers first.
- For dates, MIN returns the earliest date. Format the result cell as a date to read it.
- If the range has no numbers at all, MIN returns 0.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as leaving out the last row. **Fix:** Check the highlighted range after typing the formula.

Mistake: Typing 0 for missing values, so MIN returns 0. **Fix:** Leave missing values blank, or use text such as "TBD", so MIN ignores them.

Mistake: Getting a wrong minimum because some numbers are stored as text. **Fix:** Convert them to numbers (for example with Text to Columns) and check that they right-align.

Mistake: Expecting MIN to show who has the lowest value. **Fix:** MIN returns only the number. Use a lookup to find the name (see the tip below).

8. Data Analyst Tip

MIN gives the value, not the name. To find who has it, use `=INDEX(A2:A5, MATCH(MIN(B2:B5), B2:B5, 0))`, which returns Rahim for the first example. MIN and MAX together show the spread of your data: `=MAX(B2:B5) - MIN(B2:B5)` gives 25000.